Understanding the Git Commit Graph

Explanation of the symbols and history in the supplied Git graph.

```
*   e016d70 (HEAD -> main) added markdown1
|\
| * 5946b94 (feature/add-md1) added markdown 1
* | f452741 deleted markdown1 and markdown2
* |   6642d32 Merge branch 'feature/add-md2'
|\ \
| |/
|/|
| * c0a7d39 added markdown 2
* | 2ffa582 added markdown 1
|/
* a593249 added add_num function and main script entry point
```

1. What each symbol means

Symbol	Meaning
`*`	A commit in Git history. Each asterisk marks one specific commit.
`|`	A line connecting commits along a branch's history.
`/` and `\`	Lines showing branches diverging or merging.
`HEAD`	Identifies the commit currently checked out.
`main`	A branch name pointing to a commit.
`feature/add-md1`	Another branch name pointing to a commit.
`e016d70`	The abbreviated SHA hash identifying a commit.

2. Why are there multiple asterisks?

Each * represents a different commit, not a different branch. For example:

```
*   e016d70 (HEAD -> main) added markdown1
|\
| * 5946b94 (feature/add-md1) added markdown 1
* | f452741 deleted markdown1 and markdown2
```

This shows three commits:

- `e016d70`: A commit on `main`, currently checked out.
- `5946b94`: A commit on `feature/add-md1`.
- `f452741`: An earlier commit on `main`.

The two branches have different commits at the top, so their histories have diverged.

3. What the complete history tells you

1. a593249 — Initial shared history

Added the `add_num` function and main script entry point.

2. Two branches develop

`main`: `2ffa582` — added markdown 1.
Feature branch: `c0a7d39` — added markdown 2.

3. 6642d32 — Merge commit

The `feature/add-md2` branch was merged into `main`.

4. f452741 — Delete files

A later commit deleted markdown1 and markdown2.

5. Two branches diverge again

`main`: `e016d70` — added markdown1.
`feature/add-md1`: `5946b94` — added markdown 1.

4. The most important thing to understand

A `*` does **not** mean a branch. It means a commit. A branch is a movable pointer to a commit.

```
* e016d70 (HEAD -> main)
```

This means:

- `*`: There is a commit here.
- `e016d70`: Its abbreviated commit ID.
- `HEAD -> main`: You are currently on the `main` branch, and `HEAD` points to its latest commit.

Similarly:

```
* 5946b94 (feature/add-md1)
```

This means the `feature/add-md1` branch points to that commit, separate from the current `main` tip.

One extra detail

The two latest commits have different hashes even though both messages say they added markdown 1. Git identifies commits by their content and history, not just their commit messages. Therefore, the same or similar changes can exist in separate commits with different hashes.